TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*EV Monitoring Device and Data Sharing Platform for Energy Consumption and Fleet Diagnostics*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical Materials		*Electrical/electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile/Rail Vehicles	Electrical and electronics systems	Measuring Instruments	Electronic measuring instrument
Concept of Proposed Technology	*Describe the core concept, main functions, and potential uses of the proposed technology* The core concept of the proposed technology is to enable standardized, real-world data access from electric vehicles across multiple electric vehicle (EV) brands and models, overcoming the limitations of traditional OBD systems for EVs. Unlike internal combustion engine (ICE) vehicles, EVs rely on complex high-voltage subsystem, such as battery management systems, inverters, electric motors, and thermal control, that are not covered by classic OBD-II standards. Moreover, many EV manufacturers implement proprietary data structures and extended diagnostic protocols, making generic data extraction incomplete or unreliable. New development is required for CAN FD-based UDS, Ethernet/DoIP, and overseas electric vehicles, where TNO sees value in collaborating with an SME that is specialized in OBD readout devices. The proposed solution bridges this gap by combining a multi-protocol EV data acquisition device from the EV CAN-bus with a secure data processing and analytics platform, capable of translating heterogeneous, manufacturer-specific data into standardized and Findable, Accessible, Interoperable and Reusable (FAIR) data extraction.			

	Main functions of the proposed technology: - Multi-protocol EV data access**:** support for advanced diagnostic protocols (e.g. UDS over CAN FD, Ethernet/DoIP) to access data from key EV subsystems such as the Battery Management System, power electronics, and thermal systems. - Data harmonization and abstraction: automated translation of proprietary and model-specific signals into a common data model, enabling cross-vehicle and cross-fleet comparison. - Context-aware data interpretation: automated inference of vehicle operating states (e.g. driving, charging, fast charging, idle, thermal conditioning) based on real-time data streams. - Secure data transmission and storage: encrypted data handling with controlled access for different stakeholder groups via a cloud-based platform. **Potential Uses:** - Fleet performance monitoring and optimization - Battery health and degradation analysis - Charging behavior and infrastructure planning - Driver behavior and energy efficiency feedback - Policy analysis and evidence-based EV adoption planning
Rationale for Joint R&D Project	*Describe the need for this joint R&D project and why collaboration is required* The rapid expansion and diversification of the electric vehicle (EV) market has created a growing need for reliable, real-world data to support fleet operation, charging infrastructure planning, and policy development. Existing laboratory tests and manufacturer specifications do not adequately reflect real-world vehicle performance, charging behavior, energy consumption, or battery degradation under operational fleet conditions. A major barrier is the lack of standardization among EV models in CAN-bus data available through the on-board diagnostics (OBD) connection. Differences in data structures, communication protocols, and parameter availability across vehicle models and manufacturers make large-scale, cross-fleet data collection complex, costly, and often incompatible. The proposed joint R&D project addresses this gap by developing a universal or adaptable device for consistent data extraction across multiple EV brands, combined with a secure, scalable data-sharing platform for vehicle- and fleet-level analysis. Collaboration is essential, as the challenge spans hardware development, software and data standardization, real-world fleet validation, and data governance. A joint approach integrates complementary expertise, accelerates development, and ensures that the resulting solution is robust, scalable, and market-relevant.
Key Words	*Provide key terms that describe the proposed technology and its main features (min 3, max 5)* - EV OBD Data Readout devices - Standardized Vehicle Data Interoperability - Scalable and Secure EV Fleet Data Platform

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	*Describe the specific technologies you aim to develop through this R&D project* - Remotely configurable, self-learning EV CAN-bus data acquisition device - Secure data transmission and storage to accompany this device - Automated data pre-treatment and harmonization - Context-aware data interpretation tool - Procedure for simplification, speedup and recording of CAN signal discovery and interpretation, and the organized storage of the collected knowledge, linked for use by the data acquisition device. - Map OEM-specific data into a common data model such as driving/idle/charging/fast charging/thermal conditioning - Execute long-term pilot validation and align reliability evaluation with potential customers (fleets/leasing/rent/company/governments/...)			
R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters* **Measurable R&D target performance specifications:** - Number of supported vehicle brands and platforms: 10 brands and platforms 1. Hyundai Motor Group GMP platform 2. VW Group MEB platform 3. Stellantis STLA platforms 4. BYD e-Platform 5. More to be determined based on need, market share and practical feasibility. - Number of supported vehicle models: 50 EV models - Number of supported vehicle ECUs : 10 ECUs/model*brand - Number of supported vehicle data parameters: 25 parameters/model*brand, covering: 1. Driving energy & efficiency 2. Stationary components energy & efficiency 3. Battery state & health 4. Charging sessions & behavior 5. Vehicle diagnostics & fault states - Connection success rate: 95% successful connection within 30 seconds after ignition ON - Reconnection performance: 90% automatic recovery within 10 seconds after a disconnection - Data continuity: maximum loss rate of 0.5% per vehicle operating hour. - Driving record data: collect: 100,000 EA - Vehicle fault data: collect: 1,000,000 EA - Driving context data: collect: 1,000,000 EA			
R&D Plan	The heart of the R&D work is the development of a multi-brand EV data acquisition device (WP1), that can be remotely updated and/or uses a centralized database with parameter information that allows EV signals to be extracted and interpreted over OBD. The functionality can be quickly expanded due to the tiered storage of signal interpretation data (per model, per brand, per manufacturer alliance) and tools that speed up the process of signal discovery for new models (WP5). The device (WP1) should be able to be updated to use all the collected knowledge on new models. Data is transmitted in a secure way (WP2), and filtered and harmonized across EV			

	brands/models automatically (WP3), and stored securely (WP2). Data is processed in a way that allows analysis of driver behavior and vehicle behavior, by doing some intermediate calculations and adding markers for trips, charging sessions, etc. (WP4). Practical tests can be done on at least 5 vehicles of different brands/alliances, to test the system developed in WP5, and to fill the initial database. TNO has already collected information for a number of models as well. The system is tested per component during development. Successively, in WP6 the system is tested in its entirety during a long-term validation and monitoring project, together with envisaged customers. The functionality and way of use of the acquisition device, data interpretation software and the reverse-engineering tools need to be properly documented; this is done in WP6. During development, software code is documented as well, to help further development in the future. WP6 also deals with reporting and dissemination. **Project Plan:** WP1: Development of a remotely configurable, self-learning data acquisition device with secure data transmission WP2: Setup of a secure data structure with multi-level access control WP3: Development of automated data validation, cleanup and cross-model harmonization WP4: Context-aware data interpretation tool WP5: Development of a procedure, including tools, that simplify and speed up reverse-engineering of CAN data: - Priority list of signals to be discovered, including redundant signals for validity checks - Database with existing knowledge (PIDs, data interpretation rules, translation rules for harmonization) per alliance, brand and vehicle model - Software for intelligent signal discovery for new vehicles using the database - CAN message translation and filtering device with interface WP6: Long-term pilot validation and align reliability evaluation with potential customers

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* **Technical Expertise and Research Contributions** TNO will provide its expertise and experience in extracting and interpreting EV-specific signals from electric passenger vehicles. This includes contributing to the automation and simplification of signal discovery processes, as well as advising on best practices for identifying, validating, and interpreting vehicle data. TNO can provide technical expertise from the following departments: - Sustainable Transport & Logistics (STL), focused on providing evidence based advice on electric vehicle behavior and energy consumption. - Advanced Computing Engineering (ACE), focused on developing methods and technologies for designing, building, and managing distributed data systems. **Signal Selection and Data Specifications** TNO will advise on which vehicle signals are relevant or required for specific analyses. This includes guidance on signal frequency (e.g. 1 Hz logging), data quality, accuracy, and data formats suitable for applications such as behavioral monitoring, performance analysis, basic calculations, and cross-checks. **Vehicle Selection and Research Scope** TNO will advise on the selection of vehicle families or platforms for the initial project phase, ensuring that chosen vehicles are representative, technically accessible, and aligned with the research objectives. **Knowledge sharing on hardware and software development** The TNO Sustainable Transport & Logistics Department has developed a OBD dongle based on an existing microcontroller platform and will further contribute to: - Laboratory-type approval and validation testing (hardware and software), - Further development and hardening of the LCL dongle to ensure robustness and reliability, - Ensuring compatibility with data logging requirements (e.g. continuous logging at 1 Hz). **System Integration and Data Infrastructure** TNO will support the integration of the dongle with a database or data backend, enabling structured storage, access, and analysis of collected vehicle data. **Testing, Validation, and Quality Assurance** TNO will contribute to laboratory testing, type approval-related activities, and verification of hardware and software performance to ensure compliance with technical and research standards. **Market Connections** TNO can facilitate connections with potential end users and stakeholders, such as Charge Point Operators (CPOs), fleet owners, and leasing companies, to support validation, piloting, and potential deployment of the solution. **Resources and Support** TNO will provide access to relevant technical knowledge, research infrastructure, testing facilities, and expert personnel required to support the defined project tasks and research objectives.

Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.* **Technical Contributions and Reverse Engineering** The Korean SME will be responsible for performing reverse engineering of electric vehicle (EV) signals across selected vehicle families and platforms. This includes identifying, decoding, and documenting relevant vehicle signals based on the knowledge the SME already has within OBD readouts. In close collaboration with TNO, the SME will work on automating this reverse-engineering process to improve efficiency, scalability, and repeatability over different vehicle models. **Hardware and Interface Development** The SME will design and develop a hardware device or interface that enables reliable and easy reading of EV OBD signals. This includes hardware architecture, embedded software, and communication interfaces required to access vehicle data in a standardized and secure manner. **Software and Data Infrastructure Development** The SME will develop supporting software tools and a self-built database to store, manage, and interpret decoded vehicle signals. This database will form the foundation for downstream analyses, services, and knowledge products. **Testing, Validation, and Quality Assurance** The SME will carry out functional testing and validation of the developed hardware and software solutions, including: - Verification of signal accuracy and consistency across vehicle families, - Validation of logging performance (e.g. continuous data capture at defined frequencies), - Reliability and robustness testing under real-world operating conditions. Where appropriate, testing activities will be conducted jointly with TNO or aligned with TNO's laboratory validation and type-testing activities. **Collaboration and Knowledge Exchange** The SME will actively collaborate with TNO and other project partners through regular technical meetings, joint development activities, knowledge sharing, and alignment on technical standards and research objectives. The SME will also support integration of its solutions with partner systems and contribute to joint dissemination and exploitation activities.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.* **Technological Achievements** - Multi-brand EV data acquisition device Development of a universal or modular hardware device capable of extracting standardized performance, usage, and charging data from multiple EV models and manufacturers, despite differences in OBD protocols and data structures. - Data standardization and interoperability framework Creation of a harmonized data model that translates heterogeneous vehicle data into a common format, enabling cross-fleet and cross-brand comparison of performance, efficiency, and battery health, including the implementation of a scalable, secure data sharing and diagnostics platform with controlled access. **Commercialization Potential** - Productized hardware solution Commercial deployment of the EV data acquisition device as a plug-and-play product for fleet operators, leasing companies, and service providers managing mixed EV fleets. - Data-as-a-Service (DaaS) and Software-as-a-Service (SaaS) models Subscription-based access to the data platform, analytics tools, and dashboards tailored to different user groups (e.g. fleet managers, charge point operators, policymakers). - Licensing and white-label opportunities Licensing of the data standardization framework and analytics modules to third parties such as charge point operators, OEM partners, or telematics providers. - Scalable and repeatable market entry The solution can be rolled out across regions and fleet sizes, benefiting from the accelerating adoption of EVs and increasing regulatory focus on data-driven transport planning. **Broader Industrial or Economic Impact:** - Improved EV charging infrastructure planning Data-driven insights support optimized placement and sizing of charging infrastructure, reducing investment risk and improving utilization rates. - Lower operational costs for fleets Enhanced understanding of vehicle performance and battery aging enables better fleet management, predictive maintenance, and total cost of ownership optimization. - Support for policy and standardization Real-world evidence informs EV adoption policies, incentives, and future technical standards at regional and national levels. - Acceleration of EV market maturity By reducing uncertainty around performance and infrastructure needs, the project helps accelerate EV adoption across commercial and public sectors. **Value / Potential Markets for the Innovation** - Charge Point Operators (CPOs) Access to anonymized vehicle location data combined with state-of-charge (SoC) and charging behavior enables data-driven decisions on charger placement, capacity planning, and grid integration. - EV Leasing and Fleet Management Organizations Improved insights into fleet performance, utilization patterns, and battery aging support contract optimization, residual value estimation, and risk reduction. - EV End-Users Enhanced feedback on driving behavior, energy efficiency, range impacts, and real-time

	diagnostics improves user experience and confidence in EV usage. - Governments and Public Authorities Aggregated, real-world data provides a robust evidence base for EV adoption strategies, infrastructure investment, and environmental impact assessments.

6. References

(1) Framework for Building Low-Cost OBD-II Data-Logging Systems for Battery Electric Vehicles. Ramai, Clarence, et al. 4, s.l. : Vehicles, 2022.

(2) A comprehensive methodology for developing and evaluating driving cycles for electric vehicles using real-world data. Lee, G, et al. Seoul : eTransportation, 2025, Vol. 24.

(3) R. Gijlwijk van, M. Paalvast, N. E. Ligterink en R. Smokers, "Real-world fuel consumption of passenger cars and light commercial vehicles", TNO, 2020.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization